Marwan Zein

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Summary

Computer Science student (Alexandria University, expected May 2027) focused on backend and systems software engineering, with hands-on experience across distributed backend systems, cloud infrastructure, and low-level systems programming (Rust, x86_64).

Education

Faculty of Computers and Data Science, Alexandria University **Expected May 2027**
B.Sc. Computer Science – Software Engineering & Data Science *Alexandria, Egypt*

- Relevant coursework: Data Structures & Algorithms, Machine Learning, AI, OOP, Statistics, Linear Algebra

Experience

Junior Backend Developer – Dice Marketing & Advertising **August 2026 – Present**
Spread Social Platform *Hybrid*

- Architected a cross-platform competitor intelligence pipeline spanning **7 social networks**, implementing **30+ scraping consumers** and processing **200+ social metrics** per competitor into **MongoDB** and **ClickHouse** for low-latency analytics and historical reporting.
- Optimized a production scheduler executing **2,000+ background jobs every 30 minutes**, reducing redundant queue execution, improving worker throughput, and increasing infrastructure resource utilization under sustained production workloads.

Backend Engineering Intern – Dice Marketing & Advertising **May 2026 – August 2026**
Spread Social Platform *Hybrid*

- Engineered backend services for a production-scale social media management and analytics platform, contributing **106+ commits**, **19k+ LOC**, and **40+ REST APIs** across analytics, scheduling, media management, authentication, and competitor intelligence.
- Built distributed asynchronous workflows using Laravel Queues, **Redis**, Horizon, **WebSockets**, **Ayrshare**, and **Apify**, including queue orchestration, **rate limiting**, caching, background synchronization, and real-time event delivery.

Open Source Contributions

Full list: martell0x1.github.io/open-source

uutils/coreutils **Nov 2026**
Rust reimplementation of GNU Coreutils for cross-platform use *Remote*

- Fixed `realpath` path resolution and symlink handling, enforcing ENOTDIR for non-directory components with regression tests (PR #16153, in review).
- Fixed non-UTF-8 test output and improved parameterized testing guidelines using `#[rstest]` (PRs #16161, #16050, merged).

spdlog **Oct 2026**
High-performance C++ logging library *Remote*

- Implemented thread-safe conditional, every-N and time-throttled logging macros using C++11 atomics and compare-and-swap, with Catch2 tests (PR #3695, in review).
- Fixed MSVC and CUDA build option propagation for the header-only CMake target (PR #3694, merged).

Hyper **Sep 2026**
Open-source Rust programming language and compiler *Remote*

- Added list, array and dictionary support across interpreter and compiler paths, with runtime validation.
- Optimized dictionary lookups using hash-based access and fixed heap-allocated string memory leaks in JIT and AOT runtimes.

Topcoat **Sep 2026**
Rust web routing library built on Tokio and Tower *Remote*

- Improved Tower service integration, fixing ServeDir path-prefix handling and query-string preservation; added tests and validated with Clippy.

Projects

Monitex – Event-Driven IoT Monitoring & Anomaly Detection **Apr 2026**

- Built an event-driven platform ingesting real-time ESP32 sensor telemetry over MQTT, bridged through RabbitMQ into an ASP.NET Core backend with a live Angular dashboard.

- Integrated a Python anomaly-detection model into the pipeline and cut environment setup time by **80%** with a Dockerized one-click deployment.
- **Stack:** ASP.NET Core, Rust (ESP32), RabbitMQ, MQTT, PostgreSQL, Angular, Python – GitHub

VisionGate – Smart IoT Garage Access System (AI/ML) **May 2025**

- Built a smart garage-entry system that scans license plates via camera for automated access, combining a NestJS backend, ESP32 firmware, and a Flutter mobile app.
- Built a YOLO + OpenCV + EasyOCR pipeline for plate detection and text extraction, reaching **95%** detection accuracy.
- **Stack:** NestJS, Flutter, ESP32, MQTT, YOLO, OpenCV, Docker, Azure – GitHub

Rusty-x86_64 – Bare-Metal Rust OS Kernel **Jan 2025**

- Built a bare-metal x86_64 kernel in Rust from scratch, implementing paging, custom heap allocators, and GDT/IDT interrupt handling.
- Achieved **100%** successful boot cycles on QEMU with kernel-level unit tests, reducing memory footprint **15%** via custom allocator design.
- **Stack:** Rust, x86_64, QEMU – GitHub

FileShare – Cloud File Sharing Web App **Mar 2025**

- Built a team-based cloud file-sharing app (Node.js/Express, vanilla JS) integrated with AWS S3, EC2, IAM, and VPC, supporting multi-file uploads with shareable download links.
- Applied least-privilege IAM policies and optimized AWS Free Tier usage, cutting hosting costs **25%** while maintaining **99%** uptime in testing.
- **Stack:** Node.js, Express, AWS (S3, EC2, IAM, VPC) – GitHub

Skills

Languages: C++, Java, Python, JavaScript/Node.js, Rust, PHP, HTML/CSS

Frameworks: Laravel, NestJS, Express.js, Angular, .NET/ASP.NET Core

Data & Infra: PostgreSQL, MySQL, MongoDB, Redis, Kafka, RabbitMQ, Docker, AWS (EC2/S3/IAM/VPC), Azure, Linux

Core Concepts: OOP, MVC, Microservices, REST APIs, Auth, OWASP Top 10

Achievements & Activities

ECPC Finalist (Summer 2024) – Ranked 5th overall, 2nd within faculty among 50+ teams on first participation. Certification

Coding Instructor, ElCode-FCDS (2023) – Led 10+ workshops for 150+ students, improving student competency by 30%.

Organizing Committee, ICPC-FCDS (2023) – Organized contests and training events for 200+ participants.